

5.0 CEREBRAL PALSY



Cerebral palsy is a group of disorders that affect movement and muscle tone or posture. It's caused by damage that occurs to the immature, developing brain, most often before birth.

Signs and symptoms appear during infancy or preschool years. In general, cerebral palsy causes impaired movement associated with exaggerated reflexes, floppiness or spasticity of the limbs and trunk, unusual posture, involuntary movements, unsteady walking, or some combination of these.

People with cerebral palsy can have problems swallowing and commonly have eye muscle imbalance, in which the eyes don't focus on the same object. They also might have reduced range of motion at various joints of their bodies due to muscle stiffness.

The cause of cerebral palsy and its effect on function vary greatly. Some people with cerebral palsy can walk; others need assistance. Some people have intellectual disabilities, but others do not. Epilepsy, blindness or deafness also might be present. Cerebral palsy is a lifelong disorder. There is no cure, but treatments can help improve function.

5.1 Causes of cerebral palsy

In most cases, the exact cause of cerebral palsy is unknown.

Atypical brain development or injury to the developing brain can cause cerebral palsy. The damage affects the part of the brain that controls body movement, coordination, and posture. This brain damage usually occurs before birth, but it can also happen during birth or the first years of life.

Other possible causes include:

- Head injuries as a result of a car accident, fall, or child abuse.
- Intracranial hemorrhage, or bleeding into the brain.
- Brain infections, such as encephalitis and meningitis.
- Infections acquired in the womb, such as German measles (rubella) and herpes simplex
- asphyxia neonatorum, or a lack of oxygen to the brain during labor and delivery.
- Gene mutations that result in atypical brain development
- Severe jaundice in the infant.

5.2 Cerebral palsy types

There are four types of motor effects from cerebral palsy, depending on which areas of the brain are affected:

- spasticity
- dyskinesia
- ataxia
- hypotonia

Many people have a combination, and the apparent effects can also change over a person's lifetime as a child begins to grow and attempt to do new things, such as balance while walking.

5.2.1 Spastic cerebral palsy

Spastic cerebral palsy is the most common type of cerebral palsy, affecting approximately 80 percent of people with the condition. It causes stiff muscles and exaggerated reflexes, making it difficult to walk. Many people with spastic cerebral palsy have walking abnormalities, such as unintentionally crossing their knees or making scissor-like movements with their legs. Muscle weakness and paralysis may also be present. The symptoms can affect the entire body or just one side of the body.

5.2.2 Dyskinetic cerebral palsy

People with dyskinetic cerebral palsy have trouble controlling their body movements. The condition causes involuntary, unusual movements in the arms, legs, and hands. In some cases, dyskinetic cerebral palsy also affects the face and tongue. The movements can be slow and writhing or rapid and jerky. The movements can make it difficult for the affected person to sit, walk, swallow, or talk.

5.2.3 Ataxic cerebral palsy

Ataxic cerebral palsy is the least common type of cerebral palsy. Ataxic cerebral palsy is characterized by voluntary muscle movements that often appear disorganized, clumsy, or jerky. People with this form of cerebral palsy usually have trouble with balance and coordination. They may have difficulty walking and performing fine motor functions, such as grasping objects and writing.

5.2.4 Hypotonic cerebral palsy

Hypotonic cerebral palsy causes diminished muscle tone and overly relaxed muscles. The arms and legs move very easily and appear floppy, like those of a rag doll. Babies with this type of cerebral palsy have little control over their heads and may have trouble breathing. As they grow older, they may struggle to sit up straight as a result of their weakened muscles. They can also have or experience:

- difficulty speaking
- poor reflexes
- walking abnormalities

5.2.5 Mixed cerebral palsy

Some people have a combination of symptoms of the different types of cerebral palsy. This is called mixed cerebral palsy. In most cases of mixed cerebral palsy, people experience symptoms of both spastic and dyskinetic cerebral palsy.

5.3 Classification of cerebral palsy is classified

Cerebral palsy is classified according to the Gross Motor Function Classification System (GMFCS). Researchers at Can Child Centre for Childhood Disability Research developed the GMFCS as a universal standard for determining the physical capabilities of people with cerebral palsy.

The system focuses on:

- The ability to sit.
- The capability for movement and mobility.
- Charting independence.
- The use of adaptive technology.



There are five levels of the GMFCS. Higher levels are associated with decreasing mobility.

- **Level 1 cerebral palsy:** Level 1 cerebral palsy is characterized by being able to walk without limitations.
- **Level 2 cerebral palsy:** A person with level 2 cerebral palsy can walk long distances without limitations, but they cannot run or jump. They may need assistive devices, such as leg and arm braces, when first learning to walk. They may also need to use a wheelchair to get around outside of their home.
- **Level 3 cerebral palsy:** A person with level 3 cerebral palsy can sit with little support and stand without any support. They need handheld assistive devices, such as a walker or cane, while walking indoors. They also need a wheelchair to get around outside of their home.

- **Level 4 cerebral palsy:** A person with level 4 cerebral palsy can walk with the use of assistive devices. They're able to move independently in a wheelchair, and they need some support when sitting.
- **Level 5 cerebral palsy:** A person with level 5 cerebral palsy needs support to maintain their head and neck position. They need support to sit and stand, and they may be able to control a motorized wheelchair.

5.4 Taking Care of Children with Cerebral Palsy

Taking care of children with cerebral palsy requires a multidisciplinary approach involving parents, caregivers, healthcare professionals, and therapists. Here are some tips on how to take care of children with cerebral palsy:

- **Make Exercise a Priority:** The child doesn't know they are any different to any other child, and they will want to enjoy the same activities. It is paramount that you ensure they are active from as early as possible, as this can be significant in how cerebral palsy affects them. As cerebral palsy affects movement, the more active they are from an early age, the less severe the condition may be. Some children are not able to walk with the condition, or they may struggle, but it is important to encourage them to do as much as they can. Don't give up on your child because they struggle, keep trying with every activity, and make sure they are getting a good level of exercise every day. An active lifestyle is good for everyone, but it is particularly beneficial for children with cerebral palsy.
- **Arrange Physiotherapy:** Physiotherapists are ideal for supporting a child with cerebral palsy, as they can help improve the physical ability of the child and improve problems with movement. It is a good idea to arrange physiotherapy from as young an age as possible. If the child is also having trouble with speech, you may want to seek the help and advice of a speech therapist too. It is also a good idea to try and do some physiotherapy with the child in your own time too, as although it is time consuming, it can really help support the child with their development.

- **Stay on Top of Their Care:** The child may see a number of health care providers, from pediatricians to specialists to therapists and beyond. One doctor likely will refer them to others for treatment and keep tabs on the care that they receive elsewhere. But it's wise for you to stay informed about the child's health, too. Many doctors prefer for parents and caregivers to be very involved in the care of children with CP. You'll help make health decisions, learn which treatments and therapies work, and see when a treatment is a success or has setbacks. When doctors count you as a crucial part of the team, you're more likely to feel pleased with the child's care.
- **Monitor Health:** It is natural for the child to be dealing with a range of healthcare providers, and you should make sure you consistently stay informed of the child's health and any changes you need to be aware of. It is not easy dealing with a child with cerebral palsy, there are lots of challenges along the way, and there will be difficult decisions to be made. It is much easier for everyone if you are involved in the process and that you are as committed to supporting the child as the healthcare providers.
- **Healthy Diet:** Children with cerebral palsy have weaker bones than other children, and as such, it is even more important to ensure you provide them with a healthy diet. This is the case for anyone, but particularly important for children with cerebral palsy. It is important to provide them with a diet which has plenty of nutrients to keep them healthy and fit. In some cases, children with cerebral palsy cannot eat via their mouth, and may have a GT (gastrostomy tube), but it is still possible to ensure they still have the necessary nutrients to maintain a healthy lifestyle.
- **Choosing Appropriate Seating:** Strollers, feeding chairs, tumble form sets, headrests, seating supports, and corner chairs—all of these are forms of adaptive seating that may benefit children with cerebral palsy. Many companies make different types of seating in several categories. Strollers are a universal method of seating—almost all infants and young children are wheeled about in a stroller by their parents or other caregivers. For most children with cerebral palsy who are younger than 1 or 2 years, the standard strollers that can be purchased at any department store are adequate. A 2-year-old who is having a great deal of difficulty with head control, however, and even the 1-year-old who is not able to provide any trunk support or head control, needs more support than standard strollers provide. One good temporary alternative to buying a divergent stroller or a

wheelchair is to have an insert fabricated for the existing stroller that can provide stability, head and trunk support, and more safety.

- **Bathing Chair:** Bathing is an activity that can involve a great deal of lifting and bending over in positions that can be very stressful to the caregiver's back. Before bathing a young child on a changing table, the caregiver should adjust the level of the table to the level of the caregiver's mid-abdomen. Children from age 2 to age 10 especially enjoy being bathed in warm water and are usually bathed in a bathtub. There are several different kinds of bathing chairs that provide trunk support for the child who has poor trunk control; an occupational therapist can help you select one that is appropriate for your child. The main problem in tub bathing for caregivers, however, is lifting the child into the bathtub. Before he reaches about age 10 this may be difficult but can still be managed. After age 10, it becomes more difficult.



- **Moving:** The small infant and child is primarily moved about his environment by being carried. As the child grows but does not begin to walk, caregivers must begin to protect their backs and not simply continue to carry the child until they are physically unable to do so. This requires planning for a home environment that is accessible to a wheelchair; planning should take place when the child is between the ages of 4 and 8. Another option is to use a rolling stool and push yourself, with the child on your lap, along the stool; alternatively, if the child can hold on securely to the stool, he may be pushed along alone.
- **Feeding:** Most infants are fed while being held by the caregiver. By 9 to 12 months, however, the child should be fed while sitting in a seat, with the caregiver doing the feeding sitting in front of the child. This frees both arms and allows the caregiver to position herself at a comfortable height and have an easier view of how the child is doing.

General guidelines for protecting your back;

- Always get your body as close to the child to be lifted as possible before starting to lift.
 - Whenever possible lift by bending the knees, not by bending through the back.
 - When working standing up, adjust the work surface and the child so that he is at the level of your navel (mid-abdomen), so that you don't bend over while bathing or dressing him.
 - If you are working with a child sitting (such as feeding a child), the child's face should be almost even with yours when you sit up straight. This prevents you from slouching over and putting strain on your spine
- **Making Things Easier for You and the Child through Home Modifications:** There are various ways to modify the home environment so the child will have more independence and things will be easier on you. The cost of these modifications varies from being inexpensive to very costly. The desired modifications may involve more available space and current resources than the caregiver has. Any modification must be consistent with the cognitive level of the individual for whom it is planned. In the case of individuals with significant mental retardation, the ability to be more independent in the home could remove a barrier that is a potentially useful safety net. In other words, it is

undesirable to modify the home in a way that would allow the person more freedom of movement if the person might inadvertently hurt himself or herself. Therefore, any suggestions must be evaluated in terms of the cognitive level of the child with the disability.

- Rearrange furniture in order to remove obstructions from pathways to rooms as well as to allow for a wheelchair to turn within a room. This latter requires a space at least 60 inches by 60 inches square.
- Rearrange kitchen cabinets and refrigerator to put necessary items within reach of the child. Bathroom accessories should be arranged as well. Pull-out drawers and lazy susans can increase accessibility.
- Remove plush wall-to-wall carpeting. Finish wood floors with a nonslip finish or install industrial-type carpeting.
- Widen doorways.
- Replace doorknobs with lever door handles.
- Replace entrance steps with ramps.
- Install a hinged arm support for help with toileting. A high-rise toilet is particularly helpful, as is a bidet.
- Install single mix faucets with levered handles and include an anti-scald device.
- Adjust the height of light switches and plugs to put them within reach.
- Program portable telephones to allow your child full-time access to others.
- Install an intercom system that can be used between rooms as well as at entrances.
- Replace bathtubs with wheel-in showers.
- Install an adjustable-height sink and counter with an open front in the kitchen and bathroom.
- Install an angled mirror above stove burners to allow the child a view of the contents of pots from his or her wheelchair.
- Buy a home automation system that allows your child to control the TV, intercom, and thermostat all from a central device.
- Buy a lounge chair with electrically powered positioning and lifting ability.
- Install a fire extinguisher at an accessible level.